CUSTOMER CHURN ANALYTICS & PREDICTION

7-Member Project Execution Plan

Telco Customer Churn • Kaggle • Cross-sectional Data

PROJECT	Customer Churn Analytics & Prediction
OBJECTIVE	Predict customers likely to churn and identify customer characteristics associated with churn.
MODELS	Logistic Regression • Random Forest • XGBoost
ANALYTICS	Data cleaning • EDA • model evaluation • churn patterns • interpretation
DASHBOARD	Microsoft Power BI — 3-page interactive dashboard
MEMBERS	M1 Anajli • M2 Kalpana • M3 Muskan • M4 Abhishri • M5 Khushi • M6 Aahana • M7 Aadrikaa

Fixed rule: The folder and filenames in this document are mandatory. Do not rename files or create alternate names such as EDA_final2.ipynb.

1. Fixed Project Folder & File Structure

```
customer-churn-prediction/
├── data/
│   ├── raw/
│   │   └── Telco-Customer-Churn.csv
│   └── processed/
│       └── telco_processed.csv
├── notebooks/
│   ├── 01_preprocessing.ipynb
│   ├── 02_eda.ipynb
│   ├── 03_logistic_regression.ipynb
│   ├── 04_random_forest.ipynb
│   ├── 05_xgboost.ipynb
│   └── 06_model_comparison.ipynb
├── models/
│   ├── logistic_regression.pkl
│   ├── random_forest.pkl
│   └── xgboost.pkl
```

```
├── app/
│   └── app.py
├── reports/
│   └── figures/
├── requirements.txt
└── README.md
```

2. Exact 7-Member Ownership

Member	Name	Responsibility	Exact files / folders
M1	Anajli	Data preprocessing	01_preprocessing.ipynb + data/processed/telco_processed.csv
M2	Kalpana	Data Analytics / EDA	02_eda.ipynb + reports/figures/
M3	Abhishri	Logistic Regression	03_logistic_regression.ipynb + models/logistic_regression.pkl
M4	Muskan	Random Forest	04_random_forest.ipynb + models/random_forest.pkl
M5	Aadrikaa	XGBoost	05_xgboost.ipynb + models/xgboost.pkl
M6	Aahana	Model Comparison	06_model_comparison.ipynb
M7	Khushi	App + Power BI integration	app/app.py + Power BI dashboard

3. Tools, Hardware & Workflow

Area	Tool	Use
Coding	Python + VS Code/Jupyter	All coding, analysis and ML
Data	Pandas + NumPy	Cleaning, transformation and analysis
EDA	Matplotlib + Seaborn	Charts and exploratory analysis
ML	Scikit-learn	Logistic Regression + Random Forest
ML	XGBoost	XGBoost classifier
Dashboard	Microsoft Power BI	Final Data Analytics dashboard
App	Streamlit	Simple churn prediction interface in app.py
Storage	Google Drive	Shared source of truth for handoffs

Workflow: Raw dataset → Preprocessing → EDA → Logistic Regression / Random Forest / XGBoost → Model Comparison → Churn Prediction → Power BI Dashboard + simple Streamlit app.

Hardware: The Telco dataset is small and tabular, so GPU is not required. The 3 stronger gaming/GPU laptops can be preferred for XGBoost experiments, repeated runs and final integration. The other 4 laptops are suitable for preprocessing, EDA, Logistic Regression, Random Forest and Power BI.

4. Google Drive Structure & Handoff Rules

Folder	Contents
00_Project_Master	README, meeting notes, final report and presentation
01_M1_Data_Preprocessing	01_preprocessing.ipynb + telco_processed.csv
02_M2_EDA	02_eda.ipynb + EDA figures
03_M3_Logistic_Regression	03_logistic_regression.ipynb + logistic_regression.pkl
04_M4_Random_Forest	04_random_forest.ipynb + random_forest.pkl
05_M5_XGBoost	05_xgboost.ipynb + xgboost.pkl
06_M6_Model_Comparison	06_model_comparison.ipynb + comparison outputs
07_M7_App_PowerBI	app.py + Power BI .pbix + dashboard exports
08_Shared_Data	Approved dataset versions
09_Final	Final report, PPT, dashboard, code and README

Every weekly handoff must contain: (1) exact required file(s), (2) outputs/results, and (3) a short README explaining what was done, how to run it, and what the next member needs.

M1 — Anajli

WORK PERIOD	**17 Aug – 23 Aug 2026**
DEADLINE	**23 Aug 2026**
MAIN PURPOSE	Build the clean, trustworthy base dataset used by everyone else.

What to do

- Load the Telco Customer Churn dataset and keep the original as data/raw/Telco-Customer-Churn.csv.
- Inspect rows, columns, data types, unique values, missing values and duplicates.
- Fix data-type issues, especially TotalCharges, and document the treatment.
- Remove fields not intended as predictive features, such as CustomerID, while retaining it separately if needed for reporting.
- Create a simple data dictionary and cleaning log.
- Save data/processed/telco_processed.csv.

How to do it

- Use Pandas: read_csv(), info(), describe(), isna(), duplicated(), value_counts().
- Do not train ML this week.
- Keep every cleaning decision explainable for viva.

Exact final Google Drive handoff

- notebooks/01_preprocessing.ipynb
- data/processed/telco_processed.csv• data dictionary / cleaning notes

Handoff: Member 2 uses telco_processed.csv for EDA; Members 3–6 use the approved processed data.

WEEKLY TEAM MEETING — 23 Aug 2026: EVERYONE attends. The responsible member must show the actual work, explain what was completed, demonstrate outputs, state problems and confirm what the next member needs.

M2 — Kalpana

WORK PERIOD	**24 Aug – 30 Aug 2026**
DEADLINE	**30 Aug 2026**
MAIN PURPOSE	Perform the main Data Analytics/EDA work and produce evidence-based churn insights.

What to do

- Calculate total customers, churned customers and overall churn rate.
- Analyze churn by contract, tenure, monthly charges, internet service and payment method.
- Analyze churn by senior citizen, partner, dependents and selected services.
- Create clear charts using Matplotlib/Seaborn.
- Identify at least 5 useful findings supported by the data.
- Save reusable figures under reports/figures/.

How to do it

- Use groupby(), crosstab/pivot tables and aggregation.
- Prefer a small number of clear charts over many decorative plots.
- Give every important chart a title, labels and one-line interpretation.

Exact final Google Drive handoff

- notebooks/02_eda.ipynb
- reports/figures/*
- EDA findings summary

Handoff: Member 7 uses the EDA findings to design Power BI.

WEEKLY TEAM MEETING — 30 Aug 2026: EVERYONE attends. The responsible member must show the actual work, explain what was completed, demonstrate outputs, state problems and confirm what the next member needs.

M3 — Muskan

WORK PERIOD	**31 Aug – 6 Sep 2026**
DEADLINE	**6 Sep 2026**
MAIN PURPOSE	Build the Logistic Regression baseline model.

What to do

- Use the approved preprocessing and train/test split from Member 1.
- Train Logistic Regression.
- Generate churn predictions and probabilities.
- Calculate Accuracy, Precision, Recall, F1-score and ROC-AUC.
- Create a confusion matrix and ROC curve if practical.
- Save the trained model as models/logistic_regression.pkl.

How to do it

- Use sklearn.linear_model.LogisticRegression.
- Keep tuning limited; the purpose is a clear baseline.
- Use a consistent test set and document random_state.

Exact final Google Drive handoff

- notebooks/03_logistic_regression.ipynb
- models/logistic_regression.pkl• model metrics/results

Handoff: Member 6 receives the metrics for model comparison; Member 7 receives useful results for Power BI.

WEEKLY TEAM MEETING — 6 Sep 2026: EVERYONE attends. The responsible member must show the actual work, explain what was completed, demonstrate outputs, state problems and confirm what the next member needs.

M4 — Abhishri

WORK PERIOD	**28 Sep – 4 Oct 2026**
DEADLINE	**4 Oct 2026**
MAIN PURPOSE	Build the Random Forest model and feature-importance analysis.

What to do

- Use the same preprocessing and train/test split.
- Train Random Forest.
- Use a reasonable, limited parameter search.
- Evaluate Accuracy, Precision, Recall, F1-score and ROC-AUC.
- Create a confusion matrix and feature importance.
- Save models/random_forest.pkl.

How to do it

- Use sklearn.ensemble.RandomForestClassifier.
- Keep experiments reproducible with random_state.
- Focus on useful feature importance rather than many plots.

Exact final Google Drive handoff

- notebooks/04_random_forest.ipynb
- models/random_forest.pkl
- feature importance/results

Handoff: Members 6 and 7 receive standard metrics and feature-importance outputs.

WEEKLY TEAM MEETING — 4 Oct 2026: EVERYONE attends. The responsible member must show the actual work, explain what was completed, demonstrate outputs, state problems and confirm what the next member needs.

M5 — Khushi

WORK PERIOD	**5 Oct – 11 Oct 2026**
DEADLINE	**11 Oct 2026**
MAIN PURPOSE	Build XGBoost and provide its evaluation results.

What to do

- Use the same preprocessing and train/test split.
- Train XGBoost.
- Perform focused tuning of a small number of parameters.
- Evaluate Accuracy, Precision, Recall, F1-score and ROC-AUC.
- Create a confusion matrix.
- Save models/xgboost.pkl.

How to do it

- Use xgboost.XGBClassifier.
- A GPU may be used on stronger laptops, but is not mandatory.
- Prioritize reproducibility and churn-class performance.

Exact final Google Drive handoff

- notebooks/05_xgboost.ipynb
- models/xgboost.pkl
- XGBoost metrics/results

Handoff: Member 6 receives the metrics for the final model comparison.

WEEKLY TEAM MEETING — 11 Oct 2026: EVERYONE attends. The responsible member must show the actual work, explain what was completed, demonstrate outputs, state problems and confirm what the next member needs.

M6 — Aahana

WORK PERIOD	**12 Oct – 18 Oct 2026**
DEADLINE	**18 Oct 2026**
MAIN PURPOSE	Compare all three models and produce the final model recommendation.

What to do

- Use Logistic Regression, Random Forest and XGBoost on the same test set.
- Compare Accuracy, Precision, Recall, F1-score and ROC-AUC.
- Create a clear model comparison table/chart.
- Determine the recommended model; do not choose using accuracy alone.
- Generate final predictions with customer identifier, actual churn, predicted churn and churn probability for Power BI/app integration.
- Complete the model-comparison notebook.

How to do it

- Load the saved models where appropriate.
- Keep the evaluation setup identical across models.
- Focus on a defensible recommendation for churn prediction.

Exact final Google Drive handoff

- notebooks/06_model_comparison.ipynb
- model_comparison.csv• final_predictions.csv

Handoff: Member 7 uses final_predictions.csv and model_comparison.csv for Power BI and the app.

WEEKLY TEAM MEETING — 18 Oct 2026: EVERYONE attends. The responsible member must show the actual work, explain what was completed, demonstrate outputs, state problems and confirm what the next member needs.

M7 — Aadrikaa

WORK PERIOD	**19 Oct – 25 Oct 2026**
DEADLINE	**25 Oct 2026**
MAIN PURPOSE	Build the simple prediction app, Power BI dashboard and final integration.

What to do

- Create app/app.py as a simple Streamlit churn prediction interface.
- Allow customer inputs and show churn prediction plus churn probability.
- Build Power BI Page 1: Churn Overview.
- Build Page 2: Churn Analytics.
- Build Page 3: ML Model Results.
- Add KPI cards and slicers.
- Use final_predictions.csv and approved model results; do not manually type metrics.
- Collect final code, dashboard and documentation into 09_Final.

How to do it

- Keep the Streamlit app simple and demonstration-ready.
- Keep Power BI to 3 focused pages.
- Check that filters work and dashboard numbers match the Python outputs.

Exact final Google Drive handoff

- app/app.py
- Customer_Churn_Dashboard.pbix
- dashboard screenshots/export
- final integrated folder

Handoff: Final team review, report, presentation and CV descriptions.

WEEKLY TEAM MEETING — 25 Oct 2026: EVERYONE attends. The responsible member must show the actual work, explain what was completed, demonstrate outputs, state problems and confirm what the next member needs.

5. Power BI Dashboard Specification

Page	Required content
Page 1 — Churn Overview	KPI cards: Total Customers, Churned Customers, Churn Rate, Average Tenure, Average Monthly Charges. Add churn distribution, churn by contract, internet service and payment method.
Page 2 — Churn Analytics	Churn vs Tenure, Monthly Charges, Tech Support, Online Security, Senior Citizen, Partner/Dependents. Add slicers for Contract, Internet Service, Gender, Senior Citizen and Payment Method.
Page 3 — ML Results	Model comparison for Logistic Regression, Random Forest and XGBoost; Accuracy, Precision, Recall, F1, ROC-AUC; confusion matrix and feature-importance/model-insight visuals where appropriate.

6. Final Team Rules & Submission Checklist

Use the exact filenames and folders in this document. No personal naming conventions.

Use one shared source of truth in Google Drive/Git; avoid silent personal copies.

Every notebook/script must have a clear execution order and use the approved data.

All three ML models must be compared on the same test set with the same metrics.

Do not add deep learning, extra datasets, web scraping, MySQL or unnecessary technologies unless faculty requires them.

Every active week ends with a whole-team meeting.

Final story: Data → Cleaning → EDA → ML → Model Comparison → Prediction → Power BI + App.

Final files to confirm

Telco-Customer-Churn.csv telco_processed.csv
01_preprocessing.ipynb
02_eda.ipynb
03_logistic_regression.ipynb
04_random_forest.ipynb
05_xgboost.ipynb
06_model_comparison.ipynb
logistic_regression.pkl
random_forest.pkl xgboost.pkl
app.py
reports/figures/ requirements.txt
README.md
Customer_Churn_Dashboard.pbix
Final report + presentation